

Mattia Piccinini

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Research Interests

My research develops autonomous **robotic systems** that can reason, plan, and act in uncertain real-world environments, under physical constraints and safety requirements. While AI has enabled robots to learn complex tasks from data, purely data-driven approaches often lack interpretability, require large datasets, and fail to guarantee safe, reliable behavior. I aim to bridge this gap by combining **physics-informed** knowledge and structured **optimization** principles with **machine learning**, producing planning and control methods that are **data-efficient**, adaptive, and safety-aware. I couple methodological advances with **real-world** validation on robotic platforms, targeting practical impact on the next-generation autonomous systems.

Education

- 11/2019 – 04/2024  **Ph.D. in Autonomous Systems**, University of Trento, Italy
Doctorate awarded with highest distinction (*summa cum laude*)
Ph.D. thesis: *Artificial Drivers for Online Time-Optimal Vehicle Trajectory Planning and Control* [[pdf](#)].
Supervisor: Prof. Francesco Biral
Date of the defense: 12/04/2024
- 09/2017 – 10/2019  **M.Sc. in Mechatronics Engineering**, University of Trento, Italy
Final grade: 110/110 *summa cum laude*
GPA: 30/30, 80% exams with maximum score, 60% exams passed with honors
Thesis: *Path planning and control of self-driving vehicles at the limits of handling* [[pdf](#)].
- 09/2014 – 09/2017  **B.Sc. in Industrial Engineering**, University of Trento, Italy
Final grade: 110/110 *summa cum laude*
GPA: 30/30, 90% exams with maximum score, 60% exams passed with honors
Thesis: *Degradation of glycol-based cooling fluids causing corrosion in machine tools refrigeration circuits*. The thesis was carried out in cooperation with the companies Pama S.p.A. and Rittal GmbH.
- 09/2009 – 06/2014  **Scientific High School Diploma**
Liceo Scientifico A. Rosmini, Rovereto, Italy
Final grade: 100/100

Research Experience

- 02/2026 – present  **Humboldt Post-doctoral Fellow**
Technical University of Munich, Munich, Germany
Supervisor: Prof. Johannes Betz
Type of contract: Post-doctoral Fellowship
- 12/2024 – 01/2026  **TUM Global Post-doctoral Fellow**
Technical University of Munich, Munich, Germany
Supervisor: Prof. Johannes Betz
Type of contract: Post-doctoral Fellowship

Research Experience (continued)

- 04/2024 – 11/2024  **Post-doctoral Researcher**
University of Trento, Trento, Italy
Supervisor: Prof. Francesco Biral
Type of contract: Assegno di ricerca (Italian research grant)
- 03/2022 – 06/2022  **Visiting Ph.D. Student**
Universität der Bundeswehr, Munich, Germany
Supervisor: Prof. Matthias Gerdts
- 11/2019 – 04/2024  **Ph.D. Student**
University of Trento, Trento, Italy
Supervisor: Prof. Francesco Biral
Types of contracts: 11/2019 – 01/2023: Borsa di dottorato (Italian Ph.D. scholarship);
02/2023 – 04/2024: Assegno di ricerca (Italian research grant)

Teaching and Supervision Experience

Metric	Value
N. courses taught (Master level)	1 (Univ. of Trento, Italy) + 3 (TUM, Germany)
N. courses taught (Ph.D. level)	1 (Univ. of Trento, Italy & TUM, Germany)
N. hours of lectures (Master level)	50 (Univ. of Trento, Italy) + 25 (TUM, Germany)
N. hours of lectures (Ph.D. level)	14 (Univ. of Trento, Italy & TUM, Germany)
N. exam sessions prepared and corrected	6 (Univ. of Trento, Italy) + 5 (TUM, Germany)
N. co-supervised Ph.D. students	1 (Univ. of Trento, Italy) + 5 (TUM, Germany)
N. supervised visiting Ph.D. students	4 (TUM, Germany)
N. co-supervised Master students	9 (Univ. of Trento, Italy) + 8 (TUM, Germany) + 1 (HAN, The Netherlands) + 1 (EIT Digital Master School, EU)

- 06/2026  **Lecturer**, Doctoral School in Materials, Mechatronics and Systems Engineering, University of Trento, Italy
Ph.D. course *Physics-informed Machine Learning for Modeling, Planning, Control and Estimation of Physical Systems*.
 - Lecturer for all lectures (14 hours).
 - Preparation of all course materials.
 - Final exam preparation and evaluation.
- 07/2025 – present  **Lecturer**, Technical University of Munich, Germany
Master course *Autonomous Vehicles: Motion Planning and Decision Making*
 - Preparation of slides and materials for selected lectures.
 - Lecturer (20 hours).
 - Support with exam preparation and correction. **Lecturer**, Technical University of Munich, Germany
Master course *F1TENTH: Autonomous Driving Hands-on*
 - Preparation of slides and materials for selected lectures.
 - Lecturer (5 hours). **Lecturer**, Technical University of Munich, Germany
Master course *Ethical Robot Systems*
 - Support with exam preparation and correction.

Teaching and Supervision Experience (continued)

09/2021 – 02/2022

09/2020 – 02/2021

11/2019 – 02/2020

- **Lecturer**, EIT Digital Master School, University of Trento, Italy
Master course *Vehicle Dynamics, Planning and Control of Robotic Cars*.
 - Building the course from scratch for the first edition in 2019.
 - 150+ hours of materials preparation.
 - Preparation of all the lecture slides and assignments.
 - Holding lectures and exercise sessions (50 hours).
 - Grading assignments and providing feedback to students.

03/2025 – 03/2026

- **Program Assistant**, Robotics Institute Germany, Germany
Assistance in the creation of a new Master of Science in AI-based Robotics across different universities in Germany.
 - Definition of requirements from the industry and research community.
 - Survey-based analysis of strengths and weaknesses of the current academic educational landscape in robotics.

2025 – present

- **Thesis Co-Supervisor**, Technical University of Munich, Germany
 - 5 Ph.D. Students
 - 7 Master Students
 - 4 Visiting Ph.D. Students

2019 – present

- **Thesis Co-Supervisor**, University of Trento, Italy
 - 1 Ph.D. Student
 - 9 Master Students

2021 – 2022

- **Thesis Co-Supervisor**, EIT Digital Master School (EU)
 - 1 Master Student
 - 1 Term Project

2020 – 2021

- **Thesis Co-Supervisor**, HAN University of Applied Sciences, Netherlands
 - 1 Master Student

Work Experience

09/2018 – 06/2019

- **Control and Simulation Engineer (Student)**
Formula SAE Project, Dynamics and Modeling Team, University of Trento, Italy
 - Development of vehicle multibody models to validate vehicle controllers.
 - Parameter optimization for the mechatronic powertrain.

09/2017 – 07/2018

- **Mechanical Design Engineer (Student)**
Formula SAE Project, Powertrain Design Team, University of Trento, Italy
 - Responsible for the mechanical transmission unit.

06/2016 – 08/2017

- **Cooling System Engineer (Student)**
Formula SAE Project, Cooling System Design Team, University of Trento, Italy
 - Design and manufacturing of the cooling system for an electric FSAE vehicle.

08/2018 – 09/2018

- **Engineering Internship**
Pama S.p.A., Rovereto (Italy)
 - Design of innovative solutions for the kinematic chain in a large-scale, high-precision milling machine divider.

09/2016

- **Engineering Internship**
Pama S.p.A., Rovereto (Italy)
 - Experimental setup to investigate corrosion phenomena arising from the degradation of glycol-based cooling fluids within machine tool circuits.

Work Experience (continued)

02/2016

09/2015

02/2015

Engineering Internship

Pama S.p.A., Rovereto (Italy)

- Design and analysis of mechanical and hydraulic assemblies for large-scale high-precision milling and boring machine tools.

Publications and Patents

Metric	Value
N. journal papers	18 (4 of which are under review)
N. conference papers	24 (2 of which are under review)
N. patents	2 (1 pending)
N. citations (Google Scholar)	445+
h-index (Google Scholar)	12

Peer-Reviewed Journal Papers

- 1 M. Piazza, A. Langmann, F. Biral, J. Betz, and M. Piccinini, "Beyond the apex: Online minimum-time velocity planning on three-dimensional paths with variable friction," *IEEE Robotics and Automation Letters*, vol. 11, no. 8, pp. 10 034–10 041, 2026. [DOI: 10.1109/LRA.2026.3707351](#)
- 2 A. Mungiello, F. Jahncke, S. Santini, J. Betz, G. P. R. Papini, and M. Piccinini, "Vehicle dynamics learning from physics priors with model-structured neural networks," *IEEE Open Journal of Intelligent Transportation Systems*, vol. 7, pp. 1181–1196, 2026. [DOI: 10.1109/OJITS.2026.3685078](#)
- 3 S. Goblirsch, M. Piccinini, J. Betz, and M. Lienkamp, "Multidirectional gaussian-process tire models for kalman filtering in vehicle dynamics state estimation," *Vehicle System Dynamics*, pp. 1–29, 2026. [DOI: 10.1080/00423114.2026.2665407](#)
- 4 A. Langmann, S. Kohl, L. Ögretmen, M. Piccinini, and J. Betz, "A multi-stage time-variant motion planner for agile autonomous driving maneuvers," *IEEE Open Journal of Intelligent Transportation Systems*, vol. 7, pp. 1136–1150, 2026. [DOI: 10.1109/OJITS.2026.3686916](#)
- 5 Y. Gao, M. Piccinini, Y. Zhang, D. Wang, K. Moller, R. Brusnicki, B. Zarrouki, A. Gambi, J. F. Totz, K. Storms, S. Peters, A. Stocco, B. Alrifae, M. Pavone, and J. Betz, "Foundation models in autonomous driving: A survey on scenario generation and scenario analysis," *IEEE Open Journal of Intelligent Transportation Systems*, pp. 1–1, 2026. [DOI: 10.1109/OJITS.2026.3660686](#)
- 6 M. Piazza, M. Piccinini, S. Taddei, F. Biral, and E. Bertolazzi, "Real-time velocity profile optimization for time-optimal maneuvering with generic acceleration constraints," *IEEE Robotics and Automation Letters*, vol. 11, no. 2, pp. 1674–1681, 2026. [DOI: 10.1109/LRA.2025.3643297](#)
- 7 F. Jahncke, B. Zarrouki, M. Piccinini, J. D'sa, D. Isele, S. Bae, and J. Betz, "Differentiable weights-varying nonlinear mpc via gradient-based policy learning: An autonomous vehicle guidance example," *IEEE Robotics and Automation Letters*, vol. 11, no. 3, pp. 3724–3731, 2026. [DOI: 10.1109/LRA.2026.3662644](#)
- 8 M. Piccinini, S. Taddei, E. Pagot, E. Bertolazzi, and F. Biral, "How optimal is the minimum-time manoeuvre of an artificial race driver?" *Vehicle System Dynamics*, vol. 63, no. 12, pp. 2213–2240, 2025. [DOI: 10.1080/00423114.2024.2407176](#)
- 9 M. Piccinini, M. Zumerle, J. Betz, and G. Pietro Rosati Papini, "A road friction-aware anti-lock braking system based on model-structured neural networks," *IEEE Open Journal of Intelligent Transportation Systems*, vol. 6, pp. 522–536, 2025. [DOI: 10.1109/OJITS.2025.3563347](#)

- 10 M. Piccinini, S. Gottschalk, M. Gerdt, and F. Biral, "Computationally efficient minimum-time motion primitives for vehicle trajectory planning," *IEEE Open Journal of Intelligent Transportation Systems*, vol. 5, pp. 642–655, 2024. [DOI: 10.1109/OJITS.2024.3476540](#)
- 11 M. Piccinini, S. Taddei, M. Larcher, M. Piazza, and F. Biral, "A physics-driven artificial agent for online time-optimal vehicle motion planning and control," *IEEE Access*, vol. 11, pp. 46 344–46 372, 2023. [DOI: 10.1109/ACCESS.2023.3274836](#)
- 12 M. D. Lio, M. Piccinini, and F. Biral, "Robust and sample-efficient estimation of vehicle lateral velocity using neural networks with explainable structure informed by kinematic principles," *IEEE Transactions on Intelligent Transportation Systems*, pp. 1–15, 2023. [DOI: 10.1109/TITS.2023.3303776](#)
- 13 M. Piccinini, M. Larcher, E. Pagot, D. Piscini, L. Pasquato, and F. Biral, "A predictive neural hierarchical framework for on-line time-optimal motion planning and control of black-box vehicle models," *Vehicle System Dynamics*, vol. 61, no. 1, pp. 83–110, 2023. [DOI: 10.1080/00423114.2022.2035776](#) eprint: <https://doi.org/10.1080/00423114.2022.2035776>.
- 14 E. Pagot, M. Piccinini, E. Bertolazzi, and F. Biral, "Fast planning and tracking of complex autonomous parking maneuvers with optimal control and pseudo-neural networks," *IEEE Access*, vol. 11, pp. 124 163–124 180, 2023. [DOI: 10.1109/ACCESS.2023.3330431](#)

Peer-Reviewed Journal Papers *Under Review*

- 1 T. Screiter, M. Piccinini, J. Dittes, L. Küssel, J. Betz, and A. J. Lilienthal, "Robotics master's education in germany: National landscape and coordination pathways," *IEEE Access*, 2026, under review.
- 2 Z. Li, F. Zhao, M. Piccinini, B. Zarrouki, Y. Gao, Z. Shan, J. Betz, C. Lv, and L. Xie, "Sgtp: Sampling-based game-theoretic planning for real-time multi-vehicle autonomous racing," *IEEE Robotics and Automation Letters*, 2026, under review.
- 3 Z. Li, B. Zhou, M. Piccinini, C. Hu, B. Zarrouki, R. Mengharam, and L. Xie, "Evo-mpcc: Enhanced velocity optimization with learning-based auto-tuning for real-time vehicle trajectory planning," *Control Engineering Practice*, 2026, under review.
- 4 D. Wang, R. Brusnicki, Z. Lai, Y. Wu, M. Piccinini, R. Yang, W. Li, and J. Betz, "Extrospective prediction for autonomous driving in emergency cut-in scenarios," *IEEE Transactions on Intelligent Transportation Systems*, 2025, under review.

Peer-Reviewed Conference Papers

- 1 M. Piccinini, P. Zambiasi, A. Mungliello, M. Piazza, F. Jahncke, and J. Betz, "Trajectory planning and control near the limits: An open experimental benchmark on the roboracer platform," in *2026 IEEE 29th International Conference on Intelligent Transportation Systems (ITSC)*, accepted, 2026.
- 2 D. Wang, H. Ye, Z. Liang, Z. Sun, Z. Lu, Y. Zhang, Y. Zhao, Y. Gao, M. Seegert, F. Schäfer, H. Qin, W. Li, L. Palmieri, F. Jahncke, M. Piccinini, and J. Betz, "Target-bench: Can video world models achieve mapless path planning with semantic targets?" In *19th European Conference on Computer Vision (ECCV)*, accepted, 2026.
- 3 F. R. Schäfer, Y. Gao, D. Wang, T. Stauner, S. Günnemann, M. Piccinini, S. Schmidt, and J. Betz, "Egodyn-bench: Evaluating ego-motion understanding in vision-centric foundation models for autonomous driving," in *19th European Conference on Computer Vision (ECCV)*, accepted, 2026.
- 4 Y. Gao, W. Miao, M. Piccinini, H. Wang, Q. Song, and J. Betz, "Chatzscenic: An iterative rag-based framework for scenario generation in autonomous driving," in *2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, accepted, 2026.
- 5 K. Moller, R. Stroop, M. Piccinini, A. Langmann, and J. Betz, "Learning to sample: Reinforcement learning-guided sampling for autonomous vehicle motion planning," in *2026 IEEE 29th International Conference on Intelligent Transportation Systems (ITSC)*, accepted, 2026.

- 6 A. Langmann, F. P. de Araujo, M. Piccinini, and J. Betz, "A hybrid sampling-based trajectory planner with game-theoretic guidance for autonomous racing," in *2026 IEEE 29th International Conference on Intelligent Transportation Systems (ITSC)*, accepted, 2026.
- 7 G. Jank, M. Piccinini, S. Wenk, P. Pitschi, J. Betz, and B. Lohmann, "Benchmarking empirical and learning-based approaches for feedforward steering control in autonomous racing," in *2026 IEEE 29th International Conference on Intelligent Transportation Systems (ITSC)*, accepted, 2026.
- 8 A. Schwehn, A. Langmann, M. Piccinini, and J. Betz, "Control-informed constraint adaptation in minimum-time trajectory planning for autonomous racing," in *2026 IEEE 29th International Conference on Intelligent Transportation Systems (ITSC)*, accepted, 2026.
- 9 R. Brusnicki, M. Piccinini, and J. Betz, "How well do vision-language models understand sequential driving scenes? a sensitivity study," in *2026 IEEE 29th International Conference on Intelligent Transportation Systems (ITSC)*, accepted, 2026.
- 10 R. Brusnicki, D. Pop, Y. Gao, M. Piccinini, and J. Betz, "Can vlms unlock semantic anomaly detection? a framework for structured reasoning," in *2026 IEEE 29th International Conference on Intelligent Transportation Systems (ITSC)*, accepted, 2026.
- 11 Y. Gao, M. Piccinini, R. Brusnicki, Y. Zhang, and J. Betz, "Nurisk: A visual question answering dataset for agent-level risk assessment in autonomous driving," in *2026 IEEE International Conference on Robotics and Automation (ICRA)*, accepted, 2026.
- 12 M. Piccinini, S. Taddei, J. Betz, and F. Biral, "Kineto-dynamical planning and accurate execution of minimum-time maneuvers on three-dimensional circuits," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 1–7. [DOI: 10.1109/ICRA55743.2025.11127446](https://doi.org/10.1109/ICRA55743.2025.11127446)
- 13 M. Kaufeld, M. Piccinini, and J. Betz, "Mp-rbfn: Learning-based vehicle motion primitives using radial basis function networks," in *2025 IEEE 28th International Conference on Intelligent Transportation Systems (ITSC)*, 2025, pp. 1262–1269. [DOI: 10.1109/ITSC60802.2025.11423753](https://doi.org/10.1109/ITSC60802.2025.11423753)
- 14 Y. Gao, M. Piccinini, K. Moller, A. Alanwar, and J. Betz, "From words to collisions: Llm-guided evaluation and adversarial generation of safety-critical driving scenarios," in *2025 IEEE 28th International Conference on Intelligent Transportation Systems (ITSC)*, 2025, pp. 2134–2141. [DOI: 10.1109/ITSC60802.2025.11423486](https://doi.org/10.1109/ITSC60802.2025.11423486)
- 15 M. Piccinini, A. Mungiello, G. Jank, G. P. R. Papini, F. Biral, and J. Betz, "Model-structured neural networks to control the steering dynamics of autonomous race cars," in *2025 IEEE 28th International Conference on Intelligent Transportation Systems (ITSC)*, 2025, pp. 4129–4136. [DOI: 10.1109/ITSC60802.2025.11423721](https://doi.org/10.1109/ITSC60802.2025.11423721)
- 16 F. Werner, S. Sagmeister, M. Piccinini, and J. Betz, "A quasi-steady-state black box simulation approach for the generation of g-g-g-v diagrams," in *2025 IEEE Intelligent Vehicles Symposium (IV)*, 2025, pp. 2503–2509. [DOI: 10.1109/IV64158.2025.11097491](https://doi.org/10.1109/IV64158.2025.11097491)
- 17 B. Zhou, B. Zarrouki, M. Piccinini, C. Hu, L. Xie, and J. Betz, "Safe reinforcement learning with a predictive safety filter for motion planning and control: A drifting vehicle example," in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025, pp. 4196–4203. [DOI: 10.1109/IROS60139.2025.11246272](https://doi.org/10.1109/IROS60139.2025.11246272)
- 18 S. Taddei, M. Piccinini, and F. Biral, "Biasing the driving style of an artificial race driver for online time-optimal maneuver planning," in *2025 IEEE Intelligent Vehicles Symposium (IV)*, 2025, pp. 640–647. [DOI: 10.1109/IV64158.2025.11097381](https://doi.org/10.1109/IV64158.2025.11097381)
- 19 S. Gottschalk, M. Gerdts, and M. Piccinini, "Reinforcement learning and optimal control: A hybrid collision avoidance approach," in *Proceedings of the 10th International Conference on Vehicle Technology and Intelligent Transport Systems - VEHITS, INSTICC, SciTePress*, 2024, pp. 76–87, ISBN: 978-989-758-703-0. [DOI: 10.5220/0012569800003702](https://doi.org/10.5220/0012569800003702)

- 20 M. Piazza, M. Piccinini, S. Taddei, and F. Biral, "Mptree: A sampling-based vehicle motion planner for real-time obstacle avoidance," 10, 17th IFAC Symposium on Control of Transportation Systems CTS 2024, vol. 58, 2024, pp. 146–153. [DOI: https://doi.org/10.1016/j.ifacol.2024.07.332](https://doi.org/10.1016/j.ifacol.2024.07.332)
- 21 M. Piccinini, S. Taddei, M. Piazza, and F. Biral, "Impacts of g-g-v constraints formulations on online minimum-time vehicle trajectory planning," 10, 17th IFAC Symposium on Control of Transportation Systems CTS 2024, vol. 58, 2024, pp. 87–93. [DOI: https://doi.org/10.1016/j.ifacol.2024.07.323](https://doi.org/10.1016/j.ifacol.2024.07.323)
- 22 E. Pagot, M. Piccinini, and F. Biral, "Real-time optimal control of an autonomous rc car with minimum-time maneuvers and a novel kineto-dynamical model," in *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2020, pp. 2390–2396. [DOI: 10.1109/IROS45743.2020.9340640](https://doi.org/10.1109/IROS45743.2020.9340640)

Peer-Reviewed Conference Papers *Under Review*

- 1 Y. Gao, S. Müller, M. Piccinini, M. Kaufeld, Y. Zhang, F. R. Schäfer, Q. Song, and J. Betz, "Plannerforge: Llm agents for scenario-based testing of motion planners in autonomous driving," in *2026 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, under review, 2026.
- 2 Y. Gao, C. Li, L. Y. Loon, F. R. Schäfer, F. Jahncke, M. Piccinini, Q. Song, S. Schmidt, and J. Betz, "Loop2scenic: Multimodal closed-loop generation of safety-critical scenarios for autonomous driving," in *Proceedings of the 41st AAAI Conference on Artificial Intelligence*, under review, 2026.

Peer-Reviewed Workshop Papers

- 1 C. Oefinger, F. R. Schäfer, K. Moller, M. Piccinini, and J. Betz, "Validate the dream before you trust its verdict: Admissibility for world-model simulators," in *Workshop Robot World Models, RSS Conference*, 2026.
- 2 M. Piazza, M. Piccinini, S. Taddei, F. Biral, and E. Bertolazzi, "Fast and flexible velocity planning for time-optimal maneuvers," in *Workshop on Frontiers of Optimization for Robotics, IEEE International Conference on Robotics and Automation (ICRA)*, 2026.
- 3 M. Piazza, M. Piccinini, S. Taddei, F. Biral, and E. Bertolazzi, "Fbga: A forward-backward method for online time-optimal velocity planning with generic acceleration constraints," in *2nd German Robotics Conference*, 2026. [URL: https://www.mos.ed.tum.de/fileadmin/w00ccp/avs/_my_direct_uploads/Mattia.pdf](https://www.mos.ed.tum.de/fileadmin/w00ccp/avs/_my_direct_uploads/Mattia.pdf)
- 4 R. Brusnicki, M. Piccinini, and J. Betz, "Bridging the loco-manipulation disconnect: Dynamic whole-body impulse control on floating-base robots," in *2nd German Robotics Conference*, 2026. [URL: https://www.mos.ed.tum.de/fileadmin/w00ccp/avs/_my_direct_uploads/Roberto.pdf](https://www.mos.ed.tum.de/fileadmin/w00ccp/avs/_my_direct_uploads/Roberto.pdf)
- 5 Y. Gao, M. Piccinini, and J. Betz, "Towards whole-body vla: A scalable data collection framework for quadrupedal mobile manipulators," in *2nd German Robotics Conference*, 2026. [URL: https://www.mos.ed.tum.de/fileadmin/w00ccp/avs/_my_direct_uploads/Yuan.pdf](https://www.mos.ed.tum.de/fileadmin/w00ccp/avs/_my_direct_uploads/Yuan.pdf)
- 6 M. Piccinini, M. Zumerle, J. Betz, and G. P. R. Papini, "Model-structured neural networks for road friction-aware anti-lock braking systems," in *Conference AUTOMATICA.IT*, 2025.
- 7 S. Taddei, M. Piccinini, and F. Biral, "From early to late apexes: Biasing the driving style of online time-optimal maneuver planning," in *Conference AUTOMATICA.IT*, 2025.
- 8 M. Piccinini, S. Gottschalk, M. Gerdt, and F. Biral, "Neural motion primitives for online time-optimal vehicle trajectory planning," in *Conference AUTOMATICA.IT*, 2024.
- 9 S. Taddei, M. Piccinini, E. Pagot, and F. Biral, "Artificial racing coach: Teaching humans how to maximize a racing vehicle's performance and drive at its limits," in *Conference AUTOMATICA.IT*, 2024.
- 10 M. Piccinini, S. Taddei, M. Larcher, M. Piazza, and F. Biral, "A physics-driven framework for online minimum-time vehicle motion planning and control," in *Conference AUTOMATICA.IT*, 2023.

- 11 E. Pagot, M. Piccinini, A. Plebe, E. Bertolazzi, and F. Biral, “Real-time autonomous parking in unstructured scenarios with an indirect optimal control approach,” in *Workshop Behavior-Driven Autonomous Driving in Unstructured Environments (BADUE), IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022. [URL: https://obj.umiacs.umd.edu/badue-accepted/8.pdf](https://obj.umiacs.umd.edu/badue-accepted/8.pdf)

Other Publications

- 1 M. Piccinini and S. Mariano, “Localization and control of autonomous vehicles in an urban intersection scenario,” 2019. [URL: https://www.researchgate.net/publication/334945954_Localization_and_Control_of_Autonomous_Vehicles_in_an_Urban_Intersection_Scenario](https://www.researchgate.net/publication/334945954_Localization_and_Control_of_Autonomous_Vehicles_in_an_Urban_Intersection_Scenario)

Invited Talks

- 1 M. Piccinini, “Differentiable Physics-Informed Learning and Control in Autonomous Vehicle Racing”, *Invited Talk*, 16th Workshop on Planning, Perception and Navigation for Intelligent Vehicles, 2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Pittsburgh, USA, 2026.
- 2 M. Piccinini, “Differentiable Learning for Planning and Control: From Small-Scale to Full-Scale Autonomous Vehicles with Autoware”, *Invited Talk*, Workshop An Introduction to Autoware and Its Application Platforms, IEEE 29th International Conference on Intelligent Transportation Systems (ITSC), Naples, Italy, 2026.
- 3 M. Piccinini, “Physics-Encoded Trajectory Planning and Control: From Racing Cars to General Robotics”, *Invited Talk*, University of Verona, 2026.
- 4 M. Piccinini, “Physics-Guided Motion Generation and Control: From Autonomous Racing Towards General Robotics”, *Invited Talk*, Eindhoven University of Technology, 2025.
- 5 M. Piccinini, “NNode: An Open Framework for Model-Structured Neural Networks in Robotics”, *Invited Talk*, Workshop series organized by the Ekumen company, 2025. [URL: https://www.youtube.com/watch?v=mz70WepnwMM](https://www.youtube.com/watch?v=mz70WepnwMM)
- 6 M. Piccinini, “Artificial Drivers to Learn the Vehicle Dynamics, Plan and Execute Online Time-Optimal Maneuvers”, *Invited Talk*, Technical University of Munich, 2024.
- 7 M. Piccinini, “Model-Structured Neural Networks to Model and Control Physical Systems”, *Invited Talk*, Seminar for PhD students, University of Trento, 2024.
- 8 M. Piccinini, “Virtual race drivers for planning and control with robotic cars”, *Invited Talk*, Universität der Bundeswehr, Munich, 2022.

Poster Sessions

- 1 M. Piccinini, A. Mungello, J. Betz, and G. P. R. Papini, “Model-Structured Neural Networks to Model & Control Robots”, *Late-Breaking Results Poster Session*, International Conference on Robotics and Automation (ICRA), Atlanta, USA, 2025.
- 2 M. Piccinini, S. Taddei, and F. Biral, “Artificial Race Driver”, *Poster Session at the Industrial Engineering Day*, University of Trento, 2023.
- 3 M. Piccinini, E. Pagot, and S. Taddei, “Virtual Driver to Control Race Cars near the Limits”, *PhDII Poster Session*, University of Trento, 2022.
- 4 M. Piazza and M. Piccinini, “MPTree: Motion Primitive Tree Exploration for Trajectory Planning with Dynamic Obstacle Avoidance”, *Poster Session at the Industrial Engineering Day*, University of Trento, 2022.

Patents

- 1 F. Jahncke, B. Zarrouki, M. Piccinini, J. D'sa, D. Isele, S. Bae, and J. Betz, "Differentiable Weights-Varying Nonlinear MPC via Gradient-Based Policy Learning: An Autonomous Vehicle Guidance Example", Patent pending, 2026.
- 2 M. D. Lio, F. Biral, and M. Piccinini, "Device and method, based on neural networks, for estimating the lateral speed of vehicles", WO2023166536A1, 2023. [URL: https://patents.google.com/patent/WO2023166536A1/en?q=PCT%2FIT2023%2F050058](https://patents.google.com/patent/WO2023166536A1/en?q=PCT%2FIT2023%2F050058)

Awards & Funding

- | | | |
|---------|---|---|
| 06/2026 | 📖 | Huawei X TUM MIRMI Joint Lab Call
1 fully-funded Ph.D. position for 24 months, following the evaluation of a research proposal. |
| 02/2026 | 📖 | Seal of Excellence for Marie Skłodowska-Curie Post-doctoral Fellowship
Project evaluation: 90.6% (Seal of Excellence). Not funded due to budget limitations. |
| 12/2025 | 📖 | Humboldt Post-doctoral Fellowship
2-year funding for post-doctoral researchers, following the evaluation of a research proposal (success rate < 20%). |
| | 📖 | Best Poster Award
German Autonomy Summit, Munich (Germany), 2025 (3 rd place). |
| 10/2025 | 📖 | Best Workshop Paper Award
2025 IEEE IROS conference, Workshop on Planning, Perception and Navigation for Intelligent Vehicles. |
| 09/2025 | 📖 | Best Ph.D. Thesis Award
Issued by the journal <i>MDPI Vehicles</i> . |
| 08/2025 | 📖 | IEEE ITSS Best Dissertation Award
Issued by the IEEE Intelligent Transportation Systems Society. The related article, published in the IEEE ITS Magazine, is available here . |
| | 📖 | IEEE IES-SYPA Travel Award
Issued by the IEEE Industrial Electronics Society (IES) for participation in the 2025 IEEE IROS conference (success rate < 10%). |
| 06/2025 | 📖 | Best Paper Award Finalist
2025 IEEE Intelligent Vehicles (IV) Symposium. |
| 03/2025 | 📖 | EuroTech Visiting Researcher Award
15-day research travel grant to collaborate with the Eindhoven University of Technology, following a competitive selection of a research proposal. |
| 12/2024 | 📖 | TUM Global Post-doctoral Fellowship
2-year funding for early-career researchers, following the evaluation of a research proposal (10 funded positions, success rate < 10%). |
| 07/2024 | 📖 | IFAC Young Author Award
Awarded at the 17 th IFAC Symposium on Control in Transportation Systems, Cyprus. |
| 05/2024 | 📖 | Best Paper Award Finalist
10 th International Conference on Vehicle Technology and Intelligent Transport Systems (VEHITS). |
| 04/2024 | 📖 | Ph.D. Graduation with Highest Distinction (<i>summa cum laude</i>) |
| 11/2023 | 📖 | Best Poster Award
Industrial Engineering Day, University of Trento (Italy). |
| 10/2019 | 📖 | M.Sc. Graduation with Highest Distinction (<i>summa cum laude</i>) |
| 09/2017 | 📖 | B.Sc. Graduation with Highest Distinction (<i>summa cum laude</i>) |

Projects

- 02/2026 – present  **Boosting Robot Performance with Model-Based and Data-Driven Control**
Project funded by the Alexander von Humboldt Foundation (Germany) as part of the Humboldt Postdoctoral Fellowship, following a competitive selection process.
Responsibilities: Project development and reporting.
- 05/2025 – present  **Modular Software Stacks for Autonomous Driving**
Industry-funded project, with the Honda Research Institute company (USA).
Responsibilities: Support with project development and reporting.
- 03/2025 – 03/2026  **Robotics Institute Germany – Education Cluster**
Funded by the German Federal Ministry of Research, Technology and Space (BMFTR)
Responsibilities: Assistance in the creation of a new Master of Science in AI-based Robotics across different universities in Germany; Definition of requirements from the industry and research community; Survey-based analysis of strengths and weaknesses of the current academic educational landscape in robotics.
- 01/2024 – 11/2024  **Digital Twin Vehicle Models for Energy Management, Performance Optimization and Parameter Estimation**
Industry-funded project, with the Stellantis company (Italian division).
Responsibilities: Project implementation and reporting.
- 11/2019 – 11/2021  **Artificial Drivers of Autonomous Race Cars**
Industry-funded project, with the AnteMotion S.r.l. company (Italy).
Responsibilities: Project implementation and reporting.

Academic Engagement

Editorial Activities

- **Associate Editor**, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- **Associate Editor**, IEEE International Conference on Intelligent Transportation Systems (ITSC)
- **Invited Session Organizer**, Invited Session on “Automated Vehicles at the Limits: Lessons and Opportunities from the Racetrack”, IEEE International Conference on Intelligent Transportation Systems (ITSC), 2026
- **Guest Editor** of the Special Issue on “Methods and Algorithms for Autonomous Systems in Intelligent Transportation”, IEEE Open Journal of Intelligent Transportation Systems, from August 2026

Conference Committees

- **Chair** of the session “Path Planning”, at the Automatica.it conference, Bolzano (Italy), 2024
- **Co-chair** of the session “Modeling, Control, and Optimization of Transportation Systems”, at the 17th IFAC Symposium on Control in Transportation Systems (CTS), Ayia Napa (Cyprus), 2024

Workshop Organization

- **Workshop Organizer**: M. Piccinini, M. Salazar, J. Betz, “Automated and Electrified Racing: Opportunities for Future Intelligent Vehicles”, IEEE International Conference on Intelligent Transportation Systems (ITSC), 2026
- **Workshop Organizer**: D. Meli, M. Piccinini, F. Trotti, A. Farinelli, J. Betz, “AI Meets Control for Resilient and Reliable Robot Planning (AIM-Ctrl)”, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026

University Service

- **TUM Internationalization** support: organized the research visits of Prof. Basilio Lenzo, Prof. Francesco Biral, Dr. Mattia Laurini, and the visiting Ph.D. students Aniello Mungiello, Gonçalo Torres, Mattia Piazza, and Gianluca Frison to the Technical University of Munich, Professorship of Autonomous Vehicle Systems (2025 – present).
- **Final Examination Committee** Member for around 35 Master's theses at the Technical University of Munich, Professorship of Autonomous Vehicle Systems (2024 – present).
- **Final Examination Committee** Member for around 20 Bachelor's theses at the Technical University of Munich, Professorship of Autonomous Vehicle Systems (2024 – present).
- **Selection Committee Chair** for external visiting Ph.D. positions and M.Sc. theses at the Technical University of Munich, Professorship of Autonomous Vehicle Systems (2025 – present).
- **Open Days** Lab Presenter, Master of Science in Mechatronics Engineering, University of Trento (2020 – 2022).

Society Memberships

- IEEE Robotics and Automation Society (RAS)
- IEEE Intelligent Transportation Systems Society (ITSS)
- Società Italiana dei Docenti e Ricercatori di Automatica (SIDRA)

Peer-Review Activities

Journals: IEEE Transactions on Intelligent Transportation Systems; IEEE Transactions on Intelligent Vehicles; IEEE Transactions on Vehicular Technology; IEEE Transactions on Mechatronics; IEEE Transactions on Industrial Electronics; IEEE Open Journal of Control Systems; IEEE Transactions on Control Systems Technology; IEEE Transactions on Robotics; IEEE Robotics and Automation Letters (RA-L); IEEE Access; Robotics and Autonomous Systems; Springer Nature - International Journal of Machine Learning and Cybernetics; Springer Optimization and Engineering; Springer Nature Communications; Control Engineering Practice (IFAC); Vehicle System Dynamics; Mechanical Systems and Signal Processing; Results in Materials (Elsevier); MDPI Vehicles; MDPI Machines; MDPI Electronics; MDPI Actuators; MDPI Algorithms; MDPI Drones; MDPI Symmetry; MDPI Applied Sciences; MDPI World Electric Vehicle Journal

Conferences: IEEE International Conference on Robotics and Automation (ICRA); IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS); IEEE Intelligent Vehicles Symposium (IV); IEEE International Conference on Intelligent Transportation Systems (ITSC); IEEE Conference on Decision and Control (CDC); Conference on Neural Information Processing Systems (NeurIPS); European Control Conference (ECC); IEEE Conference on Control Technology and Applications; IEEE International Conference on Real-time Computing and Robotics; IEEE International Conference on Advanced Motion Control; International Conference on System Theory, Control and Computing (ICSTCC); IFAC Symposium on Control in Transportation Systems; Winter Conference on Applications of Computer Vision (WACV).

Skills

Programming	■	Matlab, Simulink, Python, C/C++, Maple, Mathematica, Ruby
Machine Learning Frameworks	■	Tensorflow, Keras, PyTorch
Development Tools	■	GitHub, GitLab, Bitbucket
CAD Software	■	SolidWorks, Inventor, AutoCAD
Operating Systems	■	MacOS, Windows, Linux

Skills (continued)

Other Computer Skills	■	TeX, Microsoft Word, Microsoft Excel, Microsoft Power-Point, Apple Keynote, Apple Pages
Others	■	Driving licence B
Languages	■	English: full professional proficiency (Cambridge Certificate of Advanced English, level C1) German: upper-intermediate knowledge Italian: native

Personal Details and Interests

- Date of birth: 16/07/1995
- Nationality: Italian
- Personal interests: traveling, swimming, hiking

References

- Prof. Johannes Betz, Technische Universität München, München (Germany),
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Munich, 01/08/2026

Mattia Piccinini

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